

1.



SIMATS
ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

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QUESTIONS 48 / 48 completed

Add

ADD

Write a C++ program to add 2 matrices.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int r, c;
6     cin >> r >> c;
7
8     int a[r][c], b[r][c], sum[r][c];
9
10    for (int i = 0; i < r; i++) {
11        for (int j = 0; j < c; j++) {
12            cin >> a[i][j];
13        }
14    }
15
16    for (int i = 0; i < r; i++) {
17        for (int j = 0; j < c; j++) {
18            cin >> b[i][j];
19        }
20    }
21}
```

INPUT

2 2
1 2
3 4
5 6
7 8

2.



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Multiply

MULTIPLY

Write a C++ program to multiply 2 matrices.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int r1, c1, r2, c2;
6     cin >> r1 >> c1;
7
8     int a[r1][c1];
9     for (int i = 0; i < r1; i++) {
10         for (int j = 0; j < c1; j++) {
11             cin >> a[i][j];
12         }
13     }
14
15     cin >> r2 >> c2;
16
17     int b[r2][c2];
18     for (int i = 0; i < r2; i++) {
19         for (int j = 0; j < c2; j++) {
20             cin >> b[i][j];
21         }
22     }
23 }
```

INPUT

2 2
1 2
3 4
2 2
5 6
7 8

3.

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Diagonal

DIAGONAL

Write a C++ program
find sum of diagonals of
the matrices

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int n;
6     cin >> n;
7
8     int matrix[n][n];
9     for (int i = 0; i < n; i++) {
10         for (int j = 0; j < n; j++) {
11             cin >> matrix[i][j];
12         }
13     }
14
15     int mainDiagonal = 0;
16     int antiDiagonal = 0;
17
18     for (int i = 0; i < n; i++) {
19         mainDiagonal += matrix[i][i];
20         antiDiagonal += matrix[i][n - 1 - i];
21     }
```

INPUT

3
1 2 3
4 5 6
7 8 9